Haven: Citizen Safety and Complaint Rescue Platform

Haven Project Document

Status: Draft 1. Legal references and helpline numbers must be verified with a qualified lawyer before launch.

1. Overview

Haven lets any person file a complaint by text, voice, or photo. It covers everything from road and drainage problems to harassment, theft, threats, and disputes. An AI layer converts voice to text, summarizes the complaint, and rates urgency as Low, Medium, High, or Critical. The user immediately receives short safety steps, a plain-language explanation of their rights, and ready-to-use documents such as a complaint application or an FIR draft. The complaint reaches an admin control room, where a human reviews it, accepts it, and dispatches the right team or department. The user then tracks progress live.

Core principle: **AI suggests, humans decide.** The AI never makes the final call on urgency, dispatch, or legal action.

2. Problem Statement

- Many people, especially women, do not report harassment or threats because they are afraid, do not know their rights, or do not know the procedure.
- Civic issues such as road damage and blocked drains go unreported or unresolved because there is no clear routing to the right department.
- Many citizens cannot type well or are more comfortable speaking in Urdu, Sindhi, or English.
- When something dangerous happens, people do not know what to do in the first few minutes before help arrives.
- Filing an application or FIR is confusing, and people do not know what to write or where to go.

3. Goals and Non-Goals

Goals

1. One simple place to report any problem, with the highest focus on safety, harassment, theft, and disputes.
2. Voice complaint support in Urdu, Sindhi, and English.
3. Fast, consistent urgency rating with human review.
4. Immediate, practical safety guidance while help is on the way.
5. Legally grounded rights explanation and ready documents.
6. A control room dashboard that prioritizes critical cases and tracks every case to resolution.
7. Strong privacy protection and abuse prevention.

Non-Goals

- It does not replace the police or emergency services. It always offers one-tap calling to them.
- The AI does not give final legal judgments. All legal output is labeled as general guidance.
- It does not promise a fixed rescue time. It shows an estimated arrival time given by the dispatcher.

4. Users and Roles

Role	What they do
Citizen	Files complaints, gets guidance and documents, tracks status. Can stay anonymous to the public but verified internally.
Control Room Admin	Reviews incoming cases, confirms or changes urgency, accepts or rejects, assigns teams.
Rescue or Response Team	Receives assigned cases, shares live status and arrival time, closes the case with notes.
Department Officer	Handles civic cases such as roads, drainage, water, and sanitation.
Legal Advisor (reviewer)	Maintains and approves the legal knowledge base and document templates.

Role	What they do
Super Admin	Manages users, roles, audit logs, and settings.

5. Complaint Categories and Priority

The platform has two tracks: a Safety and Justice track (high focus) and a Civic track (lower focus).

Safety and Justice track

- Harassment (street, workplace, online, stalking)
- Threats and blackmail
- Domestic violence
- Theft, snatching, robbery
- Property and neighbor disputes
- Fraud and scams
- Missing person
- Any other personal safety issue

Civic track

- Road damage
- Drainage and sewage
- Garbage and sanitation
- Water and electricity issues
- Street lights

Urgency levels

Level	Meaning	Examples	Target handling
Critical	Life or safety in immediate danger	Someone is following me now, I am being threatened, attack in progress	Instant alert to admin, emergency call buttons shown, safety steps first
High	Serious, needs action within hours	Ongoing harassment, theft just happened, blackmail	Top of queue, human review right away

Level	Meaning	Examples	Target handling
Medium	Needs action within days	Dispute, repeated nuisance, fraud already done	Normal queue
Low	Not urgent	Pothole, broken street light, garbage pile	Routed to department, tracked

Rule: the AI assigns the level and gives its reason. Any keyword or signal for immediate danger (for example threats, weapons, "following me", "help") automatically raises the case to at least High and triggers human review within minutes.

6. End-to-End User Flow

Step 1: Open the app and choose how to report

- Big **Call 15 (Police)** and **Call 1099 (Human Rights Helpline)** buttons are always visible on every screen.
- User taps **Report a problem**.
- Optional: tap **I am in danger right now**, which skips the form and sends a quick alert with location, then continues in the background.

Step 2: Describe the problem

- Choose input: **Type**, **Speak**, or **Add photo or video**.
- Voice: the user records in Urdu, Sindhi, or English. The AI converts speech to text and shows it for the user to confirm or edit.
- The user picks a category, or leaves it to the AI to detect.
- The app collects location (GPS with manual correction), date and time, and optional details such as a description of the person or vehicle.
- The user chooses privacy: **Anonymous to the public** (identity visible only to authorized admins) or **Named**.

Step 3: AI analysis (takes seconds)

1. Speech to text (if voice).
2. Language detection and translation to a working language for the system.
3. Summary of the complaint in two to three lines.
4. Category detection.
5. Urgency rating (Low, Medium, High, Critical) with reasons.

6. Danger signal check (weapon, threat, stalking in progress, domestic violence, child involved).
7. Duplicate and spam check.

Step 4: Immediate response to the user (the key screen)

The user sees three blocks, ordered by importance:

A. Stay safe now (short, practical steps)

Generated from a reviewed safety playbook, not free-form AI text. Examples:

- Move to a well-lit, crowded place such as a shop, petrol pump, mosque, hospital, or any place with many people.
- Do not confront the person. Do not go to an isolated place.
- Stay on a call with a trusted person and share your live location with them.
- Keep your phone charged and do not delete evidence such as messages, photos, or call logs.
- If you are in a vehicle, keep doors locked and drive to a police station or crowded area.
- Tap the Call 15 button if the danger continues.
- Help is being arranged and you will see the estimated arrival time as soon as the admin accepts the case.

B. Know your rights (plain language)

- What law covers this situation, what the offence is, and what the person can legally do.
- What the police must do when a complaint or FIR is filed.
- Where to go next (police station, women protection or ombudsman office, cybercrime wing, court).
- Clearly labeled as general legal guidance and not a lawyer's opinion.

C. Your documents are ready

- Complaint application draft pre-filled with the user's details and the incident.
- FIR draft (for cases that need it, see section 8).
- Evidence checklist.
- Download as PDF, share, or print.

Step 5: Admin review and acceptance

- The case appears in the control room queue, sorted by urgency and time.
- Admin sees: summary, original voice or text, photos, location on map, AI urgency with reasons, and suggested next action.

- Admin can **Accept**, **Change urgency**, **Request more info**, **Escalate**, or **Reject with a reason**.

Step 6: Dispatch and rescue

- On acceptance, the admin assigns the nearest available team (or department for civic cases).
- The system shows the team, contact, and **estimated arrival time**, which the user sees live.
- Target for Critical cases: review within one minute and dispatch immediately, with a goal of arrival in about five minutes where a partner team is nearby. This is a service target, shown as an estimate, never as a guarantee.
- The user receives push and SMS updates: Accepted, Team assigned, Team on the way, Arrived.

Step 7: Resolution and follow-up

- The team closes the case with notes and outcome (resolved, referred to police, referred to court, no action possible).
- The user confirms whether the issue is resolved and can rate the response.
- Reopen option if the problem returns.
- Pending cases escalate automatically if not acted on within the time limit.

Civic complaints (shortened flow)

Same submission and AI analysis, but the output is a department routing (for example drainage to the water and sewerage body), a tracking number, and status updates. No emergency screens unless danger is detected.

7. Safety Guidance Engine

- Built as a **playbook library**: for each scenario (street harassment, stalking, domestic violence, theft, workplace harassment, online blackmail, and so on) a lawyer, a safety expert, or an NGO partner approves a short step list.
- The AI picks the matching playbook and adapts wording (language, location type). It does not invent new safety instructions.
- Each playbook has three parts: **Right now**, **Next hour**, **Next 24 hours**.
- Always shows the emergency buttons.
- Content is reviewed at least every six months.

8. Legal Guidance and Document Generation

Approach

- A curated **legal knowledge base** is prepared from official law texts and reviewed by a practicing lawyer. The AI only explains from this knowledge base (retrieval based), so it does not guess laws.
- Every answer shows the source law and section so it can be verified.
- A visible disclaimer on every legal screen: "General guidance based on Pakistani law. Not a substitute for a lawyer."
- A **Talk to a lawyer** option (partner lawyers or legal aid organizations) for medium, high, and critical cases.

Laws to include in the knowledge base (to be verified and kept current by the legal reviewer)

- Pakistan Penal Code (offences such as theft, robbery, criminal intimidation, assault, and insulting the modesty of a woman)
- Code of Criminal Procedure (FIR registration and police duties)
- Prevention of Electronic Crimes Act, 2016 (cyberstalking, online blackmail, and similar offences)
- Protection Against Harassment of Women at the Workplace Act, 2010
- Sindh Domestic Violence (Prevention and Protection) Act, 2013, and the equivalent law for other provinces
- Other provincial laws and local body rules for civic matters

The exact sections and recent amendments must be confirmed by the legal reviewer before they go into the app.

Documents the system generates

Document	When it is generated	Content
Complaint application	Every safety case	Addressed to the right authority, with facts, date, time, place, request for action
FIR draft or written information for FIR	High and Critical criminal cases	Structured details the police need, with relevant offence description for the officer to confirm

Document	When it is generated	Content
Harassment complaint to workplace committee or ombudsman	Workplace harassment	Based on the 2010 Act format
Cybercrime complaint	Online harassment and blackmail	Evidence list and details for the cybercrime wing
Legal notice draft (for lawyer review)	Disputes	Basic notice that a lawyer can finalize
Evidence checklist	All cases	What to save, screenshot, or photograph
Civic application	Civic cases	Department routing and tracking number

Important: the final decision about which sections of law apply is made by the police and courts. The FIR draft describes facts clearly and may suggest relevant offences, marked as "for officer confirmation".

9. AI Components

Component	Task	Human control
Speech to text	Convert Urdu, Sindhi, and English voice to text	User confirms and edits the text
Translation	Convert to the working language for analysis	Original text always kept
Summarizer	Two to three line case summary	Admin can edit
Classifier	Category and department	Admin can change
Urgency rater	Low, Medium, High, Critical with reasons	Admin confirms or overrides, and the override is logged
Danger detector	Flags immediate danger signals	Always raises priority, never lowers it
Playbook selector	Chooses the right safety steps	Uses approved content only
Rights explainer	Retrieval from the legal knowledge base	Shows sources, lawyer reviewed

Component	Task	Human control
Document drafter	Fills templates with case facts	User and lawyer can edit before use
Fake and duplicate detector	Spam and duplicate score	Admin makes the final call

Safety rules for the AI:

- The AI is never allowed to lower an urgency level on its own when danger words are present.
- If the AI is unsure, it marks the case for human review and shows safe default steps.
- If the AI service fails, the user still sees emergency buttons and the default safety steps, and the case goes to the admin queue as High priority.

10. Control Room (Admin) Design

- **Live queue** sorted by urgency, with color and sound alerts for Critical cases.
- **Case view:** summary, original message and audio, photos, map pin, user history, AI reasons, suggested action.
- **Actions:** accept, reject with reason, change urgency, assign team, request more info, escalate to police or hospital, close case.
- **Team management:** list of teams and departments with area, availability, and contact.
- **Map view:** cases and teams on a map, nearest available team suggestion.
- **SLA timers:** time to review and time to dispatch, with automatic escalation to a supervisor.
- **Audit log:** every view, change, and decision is recorded with the admin name and time.
- **Reports:** cases by category, area, response time, resolution rate, and repeat hotspots.

11. Fake Complaint and Abuse Prevention

- Phone number verification by OTP for every account.
- Rate limits per user and per device (for example a maximum number of complaints per day).
- Duplicate detection (same text, same location, and same time window).
- Trust score per user, based on past complaint outcomes. A low trust score sends cases for extra review and never blocks a Critical alert automatically.
- Location and time sanity checks.

- Admin can reject with a reason, and the user can appeal.
- Clear warning at submission that false complaints are an offence and can lead to account suspension and legal action.
- Repeated abuse leads to suspension after review by a human.

12. Privacy, Security, and Compliance

- Collect the minimum data needed.
- Anonymous mode: identity is hidden from public views and from most staff, and visible only to authorized senior admins when needed to act.
- Encryption in transit and at rest, including audio and photos.
- Role-based access, so a department officer sees only their civic cases, and only the control room sees safety cases.
- Full audit trail for every access.
- Data retention policy (for example delete audio after the case is closed and the retention period ends).
- User can request deletion of their data, except where the law requires retention.
- Handle sensitive cases (domestic violence, children) with extra restricted access.
- Disclose clearly to users how their data is used and who can see it.
- Follow applicable Pakistani data protection and cybercrime laws, and confirm with the legal reviewer.
- Do not store raw data with AI providers any longer than necessary. Use providers that allow no-training and limited retention settings.

13. System Architecture

- **Mobile app and web app:** the citizen interface (React Native or Flutter for mobile, React for web).
- **Backend API:** Node.js or Python (FastAPI) with authentication, case management, and notifications.
- **Database:** PostgreSQL (with PostGIS for location queries).
- **File storage:** encrypted object storage for audio, photos, and generated PDFs.
- **AI services:** speech to text, LLM for summary, classification, and urgency, and a retrieval layer over the legal knowledge base.
- **Realtime:** WebSockets or push services for the live admin queue and the user status updates.
- **Notifications:** push notifications, SMS gateway, and optional WhatsApp.

- **Maps:** a maps API for location, routing, and nearest team lookup.
- **Admin dashboard:** a web app for the control room.
- **Monitoring:** logs, uptime alerts, and failover for the AI service.

14. Data Model (core tables)

Table	Key fields
users	id, phone, verified, role, trust_score, language, created_at
complaints	id, user_id, category, track (safety or civic), original_text, audio_url, translated_text, summary, ai_urgency, final_urgency, urgency_reason, danger_flags, location, is_anonymous, status, created_at
attachments	id, complaint_id, type, url
playbooks	id, scenario, language, right_now, next_hour, next_24h, reviewed_by, reviewed_at
legal_entries	id, law_name, section, plain_summary, source_url, reviewed_by, reviewed_at
documents	id, complaint_id, type, content, pdf_url, created_at
teams	id, name, type, area, contact, availability
assignments	id, complaint_id, team_id, assigned_by, eta, status, arrived_at
status_updates	id, complaint_id, status, note, created_at
admin_actions	id, complaint_id, admin_id, action, reason, created_at
audit_logs	id, actor_id, action, target, created_at

15. Screens

Citizen app

1. Welcome and OTP login
2. Home (Report a problem, Call 15, Call 1099, My cases)

3. Choose input (type, speak, photo)
4. Complaint form (category, location, privacy choice)
5. AI result: urgency, summary, edit option
6. Safety steps, rights, and documents
7. Case tracking (status, team, estimated arrival)
8. Chat or call with the control room
9. My cases and downloaded documents
10. Settings and language

Control room

1. Login and role check
2. Live queue
3. Case detail
4. Map view with teams
5. Team and department management
6. Reports and analytics
7. Legal and playbook content manager
8. Audit logs

16. Scope and Roadmap

MVP (hackathon or first demo)

- Text and voice complaint submission
- Speech to text and AI summary
- AI urgency rating with reasons
- Safety steps from a small playbook set (harassment, theft, and one civic example)
- Rights explanation and one generated complaint application
- Call 15 and Call 1099 buttons
- Admin dashboard with priority queue and accept or reject
- Status updates to the user
- Basic fake complaint checks (OTP and rate limit)

Phase 2

- FIR and other document types
- Team assignment with map and estimated arrival

- Anonymous mode with role-based access
- Sindhi voice support
- SMS and WhatsApp updates

Phase 3

- Partner integrations with police, rescue services, and legal aid
- Live team tracking
- Hotspot analytics for authorities
- Trust scoring and advanced abuse detection
- Offline and low-network mode

17. Success Metrics

- Time from submission to admin review (target: under one minute for Critical)
- Time from acceptance to dispatch
- Percentage of cases where the AI urgency matches the final admin urgency
- Case resolution rate and time to resolve
- User satisfaction rating
- Share of complaints marked fake or duplicate
- Number of users who downloaded and used the generated documents

18. Risks and Mitigation

Risk	Mitigation
AI rates a dangerous case too low	Danger keyword rules override, humans review, and the override is logged
Wrong or outdated legal information	Retrieval from a lawyer-reviewed knowledge base, sources shown, and regular review
Promise of fast rescue not kept	Show estimated arrival time only, partner agreements, and fallback to Call 15
Misuse and fake complaints	OTP, rate limits, trust score, admin review, and legal warning
Privacy leak of sensitive cases	Anonymous mode, role-based access, encryption, and audit logs

Risk	Mitigation
AI or network failure	Always-visible emergency buttons and default safety steps, with cases queued as High
No partner rescue teams available	Start with referral to existing services (police, rescue, hospital) before running own teams
Legal liability for the platform	Clear disclaimers, lawyer review, and terms of use reviewed by a legal advisor

19. Partners Needed

- Police and emergency services (for case referral and response)
- Rescue and ambulance services
- Legal aid organizations and partner lawyers
- Women's rights and protection NGOs (to review playbooks)
- Local government departments for civic issues
- SMS and telecom providers

20. Open Decisions

1. Project name: Haven. Branding (logo, colors, tagline) is still to be decided.
2. Who operates the control room (own team, partner organization, or institution).
3. Which city or area for the first launch.
4. Which languages to support first.
5. Which AI providers to use for speech and language, and their data retention terms.
6. How rescue teams will be sourced and managed.
7. Which lawyer or organization will review the legal content.

21. Important Notes

- Emergency numbers used in the app: Police 15 and Ministry of Human Rights helpline 1099. Confirm these are current before launch, and consider adding other numbers (for example rescue and cybercrime helplines) after verification.
- All legal information is general guidance and must be reviewed by a qualified lawyer before release.
- The platform supports and speeds up help. It does not replace police, courts, or emergency services.